# Lucas Silva

**AI Engineer | Machine Learning | LLMs | Python**

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## SUMMARY

AI developer with 4+ years across machine learning, data engineering and full-stack development. At Siemens, builds production ML systems that automate payment-to-invoice matching for 3–5k payments a day, and leads the department's AI knowledge sharing and training. MSc in Computer Science specialised in Artificial Intelligence. Based in Portugal and open to remote roles.

## SKILLS

**AI and Machine Learning:** Machine Learning, Deep Learning, LLMs, Generative AI, Classification, Retrieval Models, PyTorch, TensorFlow, NumPy, pandas

**Programming:** Python, SQL, JavaScript, Java

**Web and APIs:** Django, REST APIs, Back-end Development, Front-end Development, Application Security

**Cloud and DevOps:** Microsoft Azure, Docker, Git, CI/CD

**Data:** Data Engineering, ETL, Data Pipelines, Data Analysis

## EXPERIENCE

**Python & AI Developer, Siemens, Lisbon, Portugal** Apr 2024 – Present

**AI payment mapping**

- Built a machine learning system that combines a classifier with a retrieval model to map incoming payments to invoices.
- Raised automated match coverage from 83–85% to about 88–90% across 3–5k payments a day, reducing manual reconciliation work. Python, PyTorch, TensorFlow, Microsoft Azure, SQL.

**Pension investment insights application**

- Full-stack development of an application that gives insight into pension investments through interactive charts and data views, supporting investment decisions.
- Responsible for application security and management. Technologies: Python, Django, REST APIs.

**AI knowledge sharing and training**

- Founded a department initiative to share AI knowledge through talks and use-case showcases, helped deliver AI trainings to colleagues, and act as a key expert and main point of contact on AI.

**Data Engineer, KPMG, Lisbon, Portugal** Sep 2022 – Mar 2024

- Advisory, Technology Consulting. Built ETL pipelines and analytics workflows in Python and SQL for client data.

**Researcher, LASIGE, University of Lisbon** 2021 – 2022

- FCT research scholarship. Deep learning (MLP, CNN, LSTM) for predicting ALS disease progression; presented at the 7th LASIGE Workshop, 2022.

**Sales Assistant, Worten, Portugal** Sep 2018 – Oct 2019

## EDUCATION

**MSc Computer Science and Engineering (AI), University of Lisbon, FCUL** 2020 – 2022
Thesis: Deep Learning for Predicting Disease Progression of Clinical Endpoints in ALS. Final grade 16/20.

**BSc Computer Science and Engineering, Lusófona University (ULHT)** 2017 – 2020

## PROJECTS

- Diabetic retinopathy detection: CNN with image preprocessing for binary and 0–4 grade classification (TensorFlow, Keras).
- Asteroids AI: genetic algorithm that evolves neural networks to play Asteroids (Python).
- Parallel temporal feature extraction: parallel feature aggregation for fraud-detection data.
- Multi-agent traffic simulation: cars and traffic lights as autonomous agents (Python, Pygame).

## CERTIFICATIONS

- Microsoft Certified: Azure Fundamentals (AZ-900); Cambridge Certificate in Advanced English (C1), 2017

## LANGUAGES

Portuguese (native), English (advanced, C1)